BA820 – Project M4

Cover Page

- **Project Title:** Hollywood Age Gap - Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 Team 10
- **Student Name:** Inchara Ashok

1. Refined Problem Statement & Focus

In this milestone, I investigate the robustness and stability of historical age gap regimes in Hollywood casting. In earlier milestones, I identified three structural "regimes" using KMeans clustering based on release year and age gap magnitude.While this three-cluster solution appeared to capture meaningful structural differences over time, visual inspection revealed some overlap between clusters, particularly in the transitional period (*Fig 1*). This raised an important methodological question - was the three-cluster solution truly justified, or would a simpler two-cluster segmentation better represent the data?

In M4, I refined my focus to - *Are the identified age gap regimes statistically and substantively robust across modeling choices and data perturbations?* Specifically, I examine robustness across a number of clusters (k), random initialization, subsampling of the dataset, and geometric validation using PCA and UMAP. This refinement shifts the analysis from exploratory segmentation to methodological validation.

Several assumptions were explicitly tested. First, I assumed that different seeds would not produce materially different cluster quality or structure. Second, because some overlap was observed in the original three-cluster solution, I questioned whether three regimes were truly justified and tested whether a two-cluster alternative might provide a cleaner or more appropriate segmentation. Third, I assumed that regime separation would be visible not only in the original feature space but also under linear (PCA) and nonlinear (UMAP) projections, indicating that the structure reflects underlying geometry rather than a modeling artifact. These assumptions were largely validated, strengthening confidence that the regime framework captures structural patterns rather than algorithmic artifacts.

2. EDA & Preprocessing: Updates

The core features used for clustering remained consistent with prior milestones: release year and age gap magnitude. Observations with missing values were removed to ensure clustering was performed on complete cases. Feature scaling (standardization) was applied before KMeans to ensure that release year and age gap contributed comparably to Euclidean distance calculations. This preprocessing choice was retained in M4 to maintain consistency across robustness tests.

No new features were added in M4. What changed is how thoroughly the clustering was validated. Cluster labels were generated for both k=2 and k=3, and a ranking system was introduced to compare clusters by their actual meaning (low, mid, high gap) rather than by their numeric labels, which can shift between runs. Subsampling was also introduced to check whether the results hold when the model is trained on a random 80% of the data. These additions didn't change the dataset but added more rigorous checks to confirm that the findings are reliable.

3. Analysis & Experiments

3.1 Baseline Regime Clustering and Validating Optimal Number of Clusters

KMeans clustering with k=3 was first applied to the standardized feature space (release year and age gap). The resulting regimes showed clear geometric separation in the original feature space - an earlier heterogeneous regime and two post-1985 regimes that differed primarily in age gap magnitude (low-gap vs high-gap) (*Fig 2*). Silhouette scores near 0.49 indicated moderate but meaningful cluster separation.

To determine the optimal number of clusters, KMeans was run for k=2 through k=10, evaluating inertia and silhouette score at each value (*Fig 3*). The silhouette score peaks at k=2 (0.503) and remains nearly as high at k=3 (0.488), before dropping sharply for k≥4. The inertia curve shows a large drop from k=2 to k=3 followed by diminishing returns, the classic elbow pattern pointing to k=3. The 0.015 silhouette difference between k=2 and k=3 is negligible. More importantly, k=2 collapses the data into a binary split that buries the most analytically valuable finding - a genuine Transition Era only surfaces at k=3. Since k=3 satisfies both validation methods and delivers meaningfully richer insights, it is retained as the optimal choice.

3.2 Random Seed Stability

To ensure clustering was not sensitive to initialization, KMeans was run across multiple random seeds [0,1,8,10,14,25,60,72,99]. The clustering solution is completely stable across all tested random seeds. Silhouette scores remain fixed at 0.488 and cluster sizes are consistent at 126, 279, and 750 movies across every run, only the numeric labels shift between runs, which is normal K-Means behavior known as label permutation (*Fig 4*). This confirms that the three regimes reflect genuine structure in the data, not a lucky initialization.

The consistent cluster sizes also allow meaningful interpretation: the 750-movie group represents the dominant modern era of relatively standardized casting, the 279-movie group captures a transition period where norms were actively shifting, and the 126-movie group corresponds to the Classical Era (roughly 1930s–1950s) where extreme gaps were most concentrated.

3.3 Geometric Validation: PCA and UMAP

Although the model uses only two features, PCA and UMAP were applied not for dimensionality reduction, but as geometric validation tools. This directly supports the core question - if regime separation only exists in Euclidean space but disappears under different geometric assumptions, the regime framework would be on shaky ground. Applying both a linear and non-linear projection provides two independent checks on the same clustering solution. PCA confirms whether the dominant axis of variation reflects a meaningful contrast between release year and

age gap under linear transformation (*Fig 5*), while UMAP tests whether separation persists under nonlinear embedding that preserves local neighborhood relationships (*Fig 6*).

The fact that the three regimes remain clearly separated in both PCA and UMAP confirms that the cluster boundaries are real, they don't just exist because of how KMeans measures distance, but reflect genuine structure in the data itself.

3.4 Robustness of Meaning (k=2 vs k=3)

This analysis focuses on whether the interpretation of regimes remains consistent across model specifications. Because the era clusters form the foundation for conclusions about inequality patterns in Hollywood casting, it is important to ensure that their meaning does not depend on a single choice of k. I compared clustering outcomes for k=2 and k=3, examining whether the relationship between age gap magnitude and rule-of-seven violations remained stable across specifications. This approach is appropriate given the simplicity of the feature space and the interpretability of cluster-level summaries such as mean age gap, violation rate, and directionality composition.

Across both k=2 and k=3, clusters were aligned by sorting them according to mean age gap (low-gap to high-gap) to avoid confusion from arbitrary numeric labels. (*Fig 7,8*) Although k=2 produced a slightly higher silhouette score, (*Fig 9,10*) the three-cluster solution preserved an intermediate gap regime that would otherwise have collapsed. Importantly, the pattern remained consistent: higher-gap regimes exhibited higher violation rates, while lower-gap regimes showed minimal violations. A key technical challenge was label switching across runs, which required alignment by gap magnitude rather than raw cluster IDs.

3.5 Subsampling Robustness

To evaluate sensitivity to data composition, I ran a subsampling robustness test with 20 independent KMeans runs (k=3), each on a random 80% subset of the data (*Fig 11*). For each run, I recorded silhouette scores, cluster sizes, and rule violation counts. This test is distinct from the seed stability check - rather than varying model parameters, it varies the data itself, providing stronger evidence that the regimes are structural patterns rather than artifacts of specific observations.

Silhouette scores remained tightly clustered between 0.48 and 0.49 across all 20 runs, and cluster size proportions were consistent up to expected label switching. The substantive pattern held in every run, one regime consistently showed high violation counts while another showed minimal violations. Minor variation in cluster sizes was observed, but no run produced a fundamentally different segmentation. This confirms that both the geometric structure and the interpretive meaning of the three regimes are stable under reasonable sampling variation.

4. Findings & Interpretations

The primary goal of this milestone was not to discover new clusters, but to validate whether the era regimes identified earlier were truly meaningful and stable. In the initial three-cluster solution, some visual overlap between regimes raised an important concern: was the segmentation genuinely capturing structural differences, or was it over-partitioning the data?. To answer this, I compared different clusters - 2 through 10, out of which 2 and 3 clusters were the most significant. Two and three-cluster models, tested stability across random initializations, and examined robustness under subsampling. Across all tests, the core structure remained intact. While the two-cluster model achieved a slightly higher silhouette score, it collapsed an important intermediate regime into a broader category. The three-cluster solution, despite minor overlap, revealed a distinct transitional regime that would otherwise be hidden. This intermediate regime is analytically valuable because it exposes a meaningful middle ground between balanced pairings and extreme inequality. The findings suggest that although k=2 is mathematically cleaner, k=3 provides richer insight into how age gap inequality evolves.

Substantively, the pattern is consistent and clear, as average age gaps increase, rule-of-seven violations increase sharply. This relationship holds whether the model uses two or three clusters and remains stable under repeated subsampling. In other words, the core inequality narrative does not depend on a specific modeling choice. The extensive validation tests therefore strengthen confidence that the regime framework reflects real structural differences rather than algorithmic artifacts.

Although the model uses only two features (release year and age gap), PCA and UMAP were applied as geometric validation tools. Their purpose was not dimensionality reduction but confirmation. PCA shows that the dominant variation in the data aligns with the historical shift in age gaps, while UMAP confirms that regime separation persists even under nonlinear transformation. The fact that regime distinctions remain visible across both linear and nonlinear projections reinforces the conclusion that the clustering captures genuine structural patterns in Hollywood casting practices.

Overall, the evidence suggests that age gap inequality in Hollywood does not decline smoothly over time. Instead, it restructures into distinct regimes, with modern casting practices splitting into both low-gap and high-gap patterns. The transitional regime identified under k=3 highlights that inequality evolves in stages rather than abruptly. By validating stability across specifications, sampling, and projection methods, this analysis confirms that the regime framework is both defensible and insightful.

Appendix

Shared GitHub Repository

- Team Github Repository - https://github.com/patelparisha/ba820-b1-team10
- Individual Github Repository - https://github.com/patelparisha/ba820-b1-team10/tree/Milestones/Inchara%20Ashok

Supplemental Material

Fig 1.

	Start Year	End Year	Movie Count	Avg Release Year
era_cluster				
1	1935	1984	126	1961.968254
2	1971	2022	279	2004.720430
0	1982	2022	750	2005.864000

[Taken from my M2 submission for reference]

Fig 2.

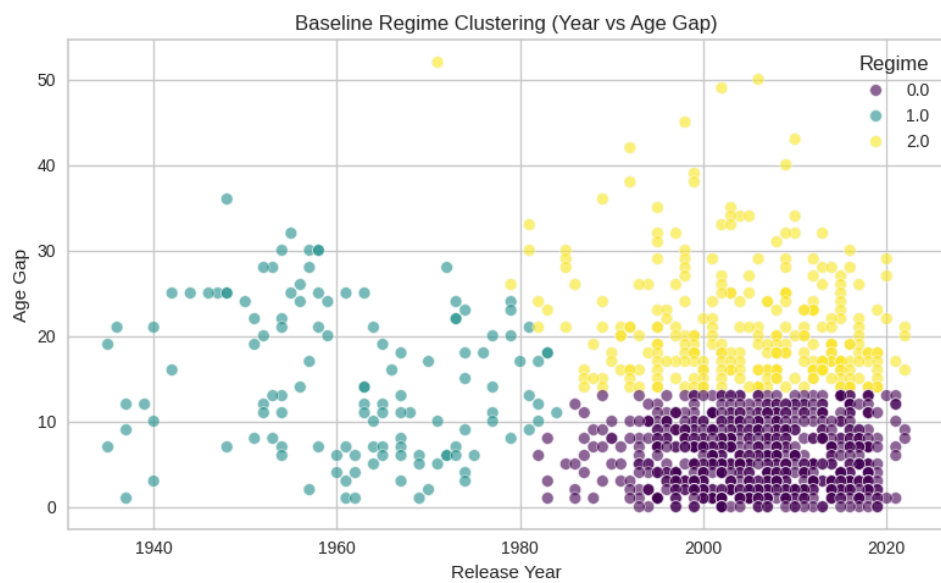


Fig 3.

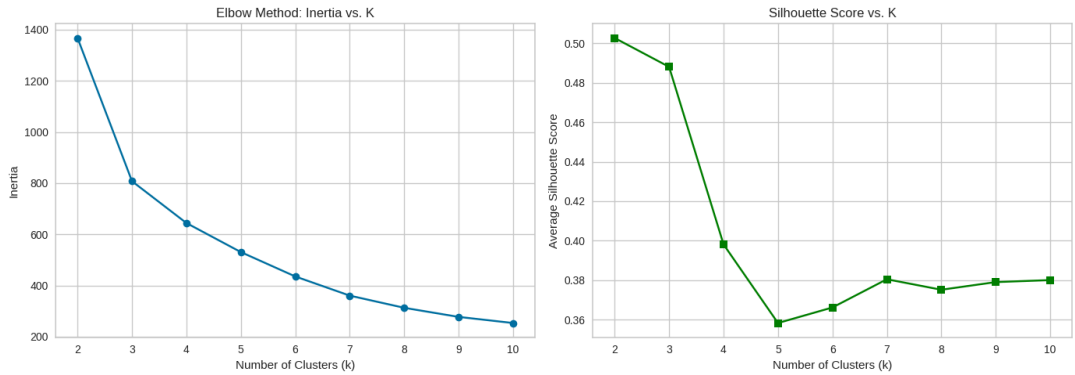


Fig 4.

Seed Stability Results:

	seed	silhouette_score	cluster_0_size	cluster_1_size	cluster_2_size
0	0	0.488086	750	126	279
1	1	0.488086	126	750	279
2	8	0.488086	126	750	279
3	10	0.488086	126	750	279
4	14	0.488086	750	126	279
5	25	0.488086	750	126	279
6	60	0.488086	279	750	126
7	72	0.488086	750	279	126
8	99	0.488086	750	126	279

Fig 5.

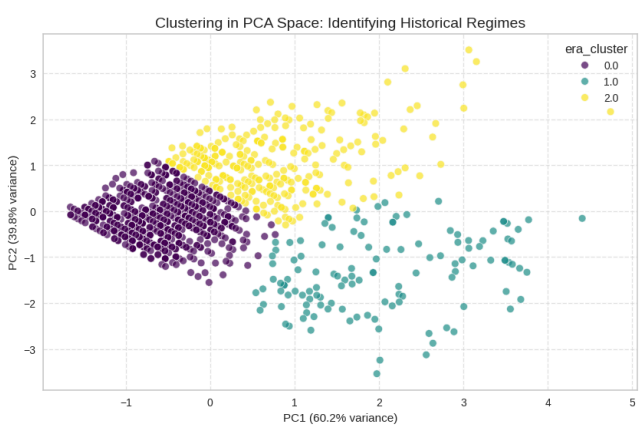


Fig 6.

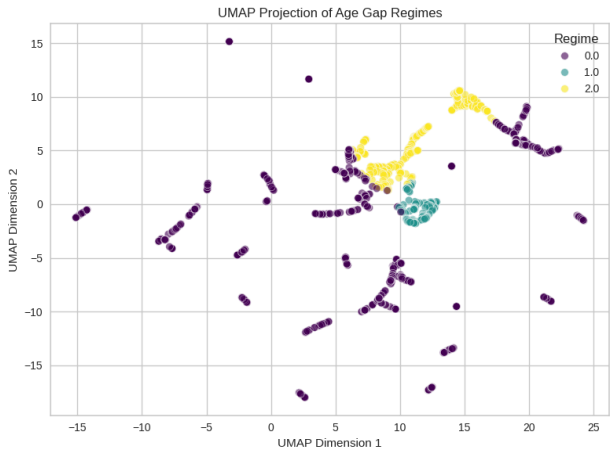
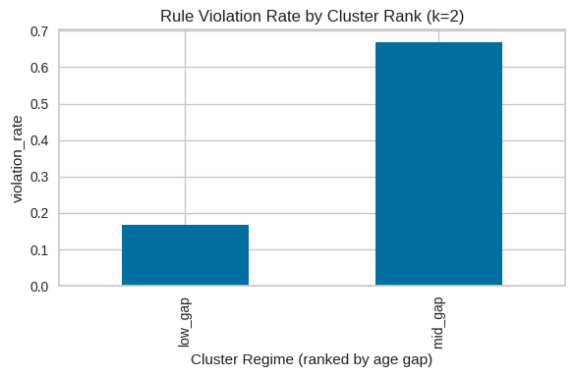


Fig 7,8.



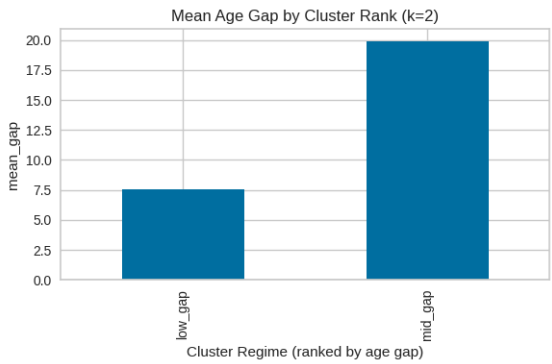
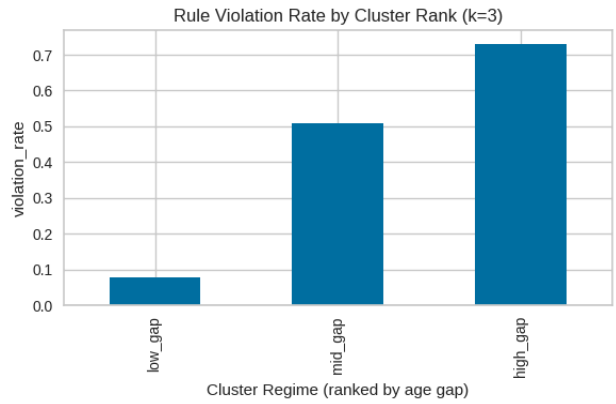


Fig 9,10.



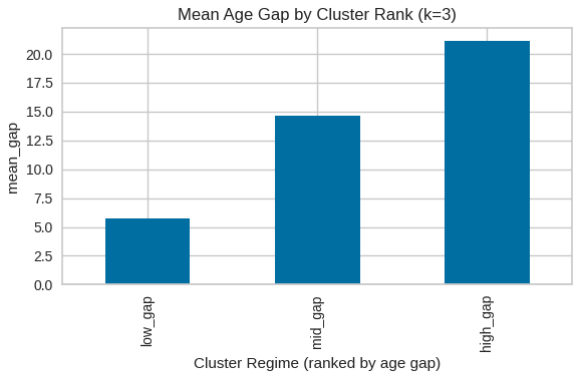


Fig 11.

	run	avg_silhouette	c0_size	c0_violations	c1_size	c1_violations	c2_size	c2_violations
0	1	0.481682	231	9	596	0	97	0
1	2	0.495087	607	0	109	1	208	8
2	3	0.479653	587	0	234	9	103	1
3	4	0.489260	101	1	594	0	229	8
4	5	0.480498	595	0	99	1	230	7
5	6	0.483420	605	0	220	6	99	1
6	7	0.488747	606	0	101	1	217	9
7	8	0.483044	94	0	599	0	231	7
8	9	0.487972	233	7	591	0	100	1
9	10	0.488009	598	0	232	4	94	1
10	11	0.485318	100	1	595	0	229	6
11	12	0.488339	599	0	106	1	219	5
12	13	0.485313	597	0	94	1	233	9
13	14	0.487879	99	0	588	0	237	6
14	15	0.494250	611	0	106	1	207	6
15	16	0.489336	103	1	597	0	224	7
16	17	0.483920	90	1	224	6	610	0
17	18	0.480617	605	0	100	1	219	5
18	19	0.480607	110	1	586	0	228	7
19	20	0.493052	595	0	228	7	101	0

Process Overview

First, preprocessing steps from prior milestones were retained, including feature selection (release year and age gap), removal of missing observations, and feature standardization to ensure balanced influence in calculations. KMeans clustering with k=3 was applied to establish the baseline regime segmentation that serves as the foundation for interpretation.

Second, a layered validation strategy was implemented to stress-test the regime structure. This included: (1) evaluating cluster quality across k=2 through k=10 using silhouette and inertia metrics, (2) testing sensitivity to random initialization across multiple seeds, (3) applying PCA and UMAP as geometric validation checks to confirm that regime separation persists under both linear and nonlinear transformations, (4) comparing k=2 and k=3 solutions to assess robustness of substantive interpretation, and (5) conducting 20 independent subsampling runs using 80% of the dataset to verify stability under data perturbation.

Use of Generative AI Tools

I used Google Gemini as a writing and structuring assistant to refine wording, reduce sections to required page limits, and organize the overall narrative. AI was also used for debugging guidance and formatting suggestions, but all analytical design, feature engineering, clustering, and interpretation were developed and implemented by myself. I used AI to understand a few concepts and to cross-verify if I was on the right path from a technical perspective. All AI-assisted text was reviewed, edited, and aligned with our actual code and results.

Link to Chatbot (AI assistance conversation used in M2, continued into M4 for consistency)